Consequences of Redundant Nodes

University of California, Davis

Author Note

Abraham Iñiguez https://orcid.org/0009-0008-5987-5107

Mijke Rhemtulla https://orcid.org/0000-0003-2572-2424

Jonathan J. Park https://orcid.org/0000-0002-6636-3504

We have no conflict of interest to disclose.

Abstract

Over the past decade, network analysis has increasingly become a popular tool for modeling psychological systems. Estimating the correct population network is critical for making we make inferences from data inferences to real-world data. However, choosing which variables or symptoms to include in a network is often challenging because the set of variables should reflect the elements of the system that the network is meant to represent. When choosing variables, it is recommended to avoid including redundant nodes. Node redundancy has previously been defined as two strongly correlated variables that share the same weighted associations with other variables (Christensen et al., 2023) ~~with other possible definitions as well~~. Given the limited research on node redundancy, how it affects estimated networks remains unclear. Plausible effects could include biasing centrality estimates, altering a network's density, or affecting its structure. Therefore, we don't presume it's a problem aim to clearly define ~~the problem of~~ node redundancy and investigate the extent to which ~~this~~ it ~~problem~~ affects network model interpretations. To address this aim, we simulated several forms of redundancy from network models of varying size. The effects of redundant nodes on ~~the~~ the network metrics were evaluated with and without ~~these~~ redundant nodes. Additionally, we assessed various forms of addressing node redundancy to see whether the ~~networks could be returned to their original~~ topology of the generating model could be recovered. Our findings provide evidence that redundant ~~models~~ nodes change the structure of estimated networks, leading to different interpretations than if redundancy were not present.

really nice flow!

Consequences of Redundant Nodes

~~Introduction~~ (Introduction is never labeled)

We use factor models to generate redundancy by making 2 indicators of one node (vs. 1 for every other node). Those 2 indicators are conditionally independent given the latent node, unlike Christensen's setup. I think we should leave LVs out of the intro...

Outline:

1. Explain redundancy and its effects are unknown in a network context

2. Here are papers that mention redundancy in some way → say what predictions/suggestions these papers have made about the effects of redundant nodes.

3. What does redundancy look like at the level of a factor model? Define a factor model. Express how "redundancy" is addressed or dealt with in the factor models [CUT]

↳ I wouldn't go here (let's talk about this)

4. How does that redundancy manifest in networks. Define networks in the psychological context.

5. Research questions which are: (1, 2, 3)

The latent variable Part is just a trick to simulate redundant nodes. I would reserve it for the method section, and keep to network models in the introduction.

We assume a linear relationship between each indicator X_i and the latent variable η in a latent variable model. This is expressed mathematically as,

$$X_i = \lambda_i \eta + \epsilon_i \tag{1}$$

where λ_i is the factor loading representing the relation between the indicator and the latent variable and ϵ_i is the measurement error. Using this linear relationship, we can obtain the correlation matrix of the indicators by,

As we see from Equation 1

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Psi}\boldsymbol{\Lambda}^T + \boldsymbol{\theta} \tag{2}$$

where Λ is the factor loading matrix of the indicators, Ψ is the correlation matrix of the latent factors, and θ is a matrix with measurement error on its diagonals. How we used this process in our simulation is described as follows:

→ end your introduction with a very clear statement of the problem, goals of this paper, and our approach.

Method

Simulation Study

We used a Monte Carlo simulation to evaluate the difference in centrality and topology between pairs of networks.

Motivating Questions (before Method)

~~A Monte Carlo simulation provides the ideal environment to assess the effect of different forms of redundancy on the surrounding nodes. Additionally, redundancy can be replicated for many iterations that is not possible with empirical datasets.~~ The first motivating question asked, what is the effect of a redundant node on node centrality metrics in a network? Centrality metrics are commonly used to describe the topology of a network in psychology. ?What does node centrality tell you about topology? Centrality measures have been argued to be ambiguous in their meaning (Bringmann et al., 2019). ok so why should we care about them?

why is this a new paragraph? Despite the criticism ~~towards~~ of centrality ~~metrics~~, the most widely applied local measures: strength, why? justify your choices! closeness, and betweenness are used here. The second motivating question asked, can the ~~original~~ not yet defined characteristics of the true network be retrieved by taking the sum score of the target and redundant node? Previous work has developed methods to detect redundancy but no such method has nobody suggested sum scores? exists on how to deal with redundancy (cite Christensen). The last motivating question asked is, can a operational definition be defined for redundancy that encompasses all forms of redundancy? the first two questions require an answer to the third - this can't really be a research question.

Method section can start here

~~Specifically, w~~We generated a "true" ~~pairs of~~ networks with ~~in which one has~~ ~~$p \times$~~ p nodes and ~~the other has~~ a "redundant" network which contained a copy of one node for a total of $p + 1$ ~~$\times p + 1$~~ ~~where the extra node is a copy of one of the nodes in the network. The network without an extra copy is termed the true network. The network with the inclusion of a node copy is termed the redundant network.~~ To do this, we simulated a random positive semi-definite correlation matrix, (which was then transformed into the true and redundant network using the latent variable model described in the introduction.) describe latent variable approach here.

+make sure enough detail (eg, R packages and methods) is given to reproduce your simulation. State where code can be found (github or OSF)

Simulation Metrics

For all of the centrality metrics discussed below, we defined an additional Δ-term for each to denote the difference between the true and redundant networks. All Δs are computed as:

$$\Delta = \cancel{G}C_{redundant} - G_{true}, \quad (3)$$

where C refers to a generic centrality metric.

In the true network, the node that is copied is called the target node. All other nodes are considered peripheral nodes labled as P1, P2, . . . , P(n − 1) . earlier you used 'p' for number of nodes. The redundant network includes all the nodes from the true network, but now also includes the copy of the target called the clone. To introduce all these terms in the previous section

~~explain~~ examine the effect of including the clone, ~~the metrics for~~ centrality of the peripheral and target nodes are the main focus in this study.

Strength

you haven't defined degree?

what is j? what is V?

it's ok to use your own letters and subscripts to be consistent with your paper!

$$\text{Strength}(v) = \sum_{j \in V} a_{vj} \tag{4}$$

Strength is similar to degree with the only difference coming from the adjacency matrix $\mathbf{A}$, where a_{vj} represents the edge weight. To compute the change in strength after excluding the clone, edges to clone are subtracted from Δ Strength to obtain Δ Strength for each node without their connection to the clone. → this can be reflected in the formula ↳ $S(v) = \sum_{j \in V'} a_{vj}$, where V' is the set of all nodes except the clone.

Closeness

$$\text{Closeness}(v) = \frac{1}{\sum_{j \in V} d(j, v)} \tag{5}$$

define terms that haven't yet been defined → what is $d(j,v)$?

Closeness measures the shortest path lengths between ~~any one~~ node v and all nodes in the network. The node with the highest closeness will generally ? take the ~~least amount~~ fewest number? of steps to each node in network.

is it not weighted?

Betweenness

define

$$\text{Betweenness}(v) = \sum_{v \neq i \neq j \in V} \frac{\sigma_{ij}(v)}{\sigma_{ij}} \tag{6}$$

Betweenness measures how often a node is a mediator on the shortest path length between two nodes. Similar to closeness, it can be used as a proxy to determine how efficiently one node can communicate with ? nodes in the network.

mediator implies causality → choose more neutral language

When computing Δ betweenness and Δ closeness excluding the clone involves removing the connections to the clone. Simply, the edges to the clone in the redundant matrix are deleted instead of subtracted as when computing Δ strength.

can you modify the formula to specify this?

Put this above the outcome metric: →

Simulation Design

Our simulation is described as follows:

describe how you obtain the true network. (LVs are for getting the clone)

(A) Simulate a random $p \times p$ correlation matrix ~~of the latent variables~~. The correlations in the matrix are sampled from a normal distribution. with what mean and variance? how did you ensure positive-definiteness?

(B) Using equation (2) produce a correlation matrix with $p + 1$ nodes. The correlation matrix from the previous step represents the matrix Ψ. The loadings in matrix Λ are chosen based on the form of redundancy. The Types of Redundancy subsection provides further details ~~on the exact loadings.~~ The measurement error in the matrix θ are values that ensure the diagonals of the final correlation matrix are one. Taken together, the resulting correlation matrix is Σ with $p + 1 \times p + 1$ nodes.

(C) Subset $p + 1 \times p + 1$ by $p \times p$ nodes ? creating two matrices that represent the true and redundant matrix. This step is done to account for measurement error ?. Thus, the resulting change from true to the redundant matrix should stem from the inclusion of a redundant node.

(D) Take the partial correlations of the true and redundant matrix, give formula and then compute the Δs for strength, closeness, and betweenness both with the clone and excluding the clone. Steps (A)-(D) are repeated for 1000 iterations. per condition

Types of Redundancy

Highly Correlated Same Function (HCSF)

This is first form of redundancy where the factor loading is set to .9 for the clone and all others nodes are set to 1. This creates a simple form of redundancy where the clone and target node form a strong connection and have similar correlations with the peripheral nodes. → target is also 1? won't this produce a clone that has no edges except to target?

Simulation Conditions

Results

Discussion

Table 1

Simulation Parameters

Condition	Value
Network Size	4, 14, 19
Correlation Strengths	NA
Signal to Noise Ratio	NA
Types of Redundancy	NA

Figure 1

scatters.png

Note.

References

Bringmann, L. F., Elmer, T., Epskamp, S., Krause, R. W., Schoch, D., Wichers, M., … Snippe, E. (2019). What do centrality measures measure in psychological networks? *Journal of abnormal psychology*, *128*(8), 892.